

Project Plan

Computer Science Department
Science and Technology School
CSC360

Title of the Project: Memory Game(or Game Suite if time allows)

Client(s): people who like to play games, possibly old people, maybe nursing home(s), or young people

Student Names:Matthew Farha, Drew Kroeger, Owen Manley

Project Overview:

The overall goal of this software project is to implement a memory game. This will make people think using short-term memory, improving or maintaining cognitive skills.

While we have not discussed the exact specifications, the “memory game” will be one of three proposals. The first and most likely proposal would be a card memory game, where the user has a large set of cards with unique icons, the cards are by default turned around, and the user can select one and try to remember where the other card with the same icon is. The other more likely proposal would be where there are three or four buttons, and you have to repeat the sequence in which the computer selected them. Each level increases the amounts of buttons pressed and buttons that must be repeated. The third and least likely proposal is the classic three-cup guessing game, which is more about eye movement than memory.

The main functionality of the software will likely include

1. multiple levels or difficulty settings: the first proposal, would include more selections of cards and icons that must chosen from, this could also involve function 2, where moves or time may be limited. The second proposal could also have more buttons involved for difficulty.
2. A timer and moves counter. This is pretty simple. The time is recorded possibly for a high score/leaderboard function. The moves counter could also be implemented for more challenges, such as limiting moves.
3. A score counter is used to keep a score and something for the user to work towards and improve.
4. Other functions are likely to be added, or subject to change. Many other unique functions could be used and added as time allows.

The languages used could vary, but will likely include some or all of the following:

Java, Python, HTML, CSS, JavaScript/TypeScript.

Libraries could include GUI tools such as Tkinter or PyQt, Version control will likely have Github/git. IDEs will be used. Graphics design will need to be done, this could include an Adobe tool like Illustrator, possibly Blender, or something much more simple like MS Paint. These,

however, are all subject to change, with many of these likely not being used, and other tools being used that are not listed.

Potential users of the project will include those who like to play simple games. Main users could include those who are trying to possibly retain or use their short-term memory, or those who are trying to determine their short-term memory size. Other users could be those who like puzzle games.

Benefits for users could include

1. cognitive enhancement or stabilization. Memory games may help improve users' memory, attention, and possibly concentration.
2. Entertainment. It is mostly a stress-free game and is not too competitive, however, with a scoring aspect, it could be made into a competitive game, which may have its own benefits.
3. Stress relief.
4. Competition: having a scoreboard can help people who are connected in one area (such as a nursing home) compete together or work together, this could also help provide social interaction.
5. Development. The user may be able to see their time or score improve over time.

Another thing I would like to add is that this is a very rough idea, and I personally (Drew) would like to come up with maybe an original type of simplistic game to add. There also may be a chance we add Connect Four- as Professor Ghazaleh requested, however, these features all depend on the smoothness of how the original project scope does. The scope will grow if we all believe it is too small, but it likely will stay the same as usually the scope grows too much.

Signatures: Drew Kroeger

Instructor Approval: